

KIRAN RAJEEV

ELECTRONICS AND COMMUNICATION ENGINEER

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PROFESSIONAL SUMMARY

Electronics and Communication Engineering graduate with hands-on exposure to PLC, SCADA, HMI, industrial networking, DCS fundamentals, and industrial communication protocols through academic projects and technical training. I am familiar with software testing concepts, Windows and Linux environments, and embedded systems fundamentals. I have a keen interest in Embedded Systems, Fiber Optic Technologies, and industrial automation, while continuously developing my technical skills through learning and upskilling. I am interested in working with a growth-oriented organization where I can apply my technical knowledge, contribute to research and product development, gain practical experience, and grow professionally.

EDUCATION

Government Engineering College, Palakkad

2021 – 2025

Bachelor of Technology in Electronics and Communication Engineering — CGPA: 8.6

Palakkad, Kerala, India

EXPERIENCE

SMEC Automation

Nov 2025 – May 2026

CRG Trainee Engineer

Cochin, India

- Assisted in PLC, SCADA and HMI development, configuration and testing for industrial automation systems.
- Performed system troubleshooting, validation and functional testing during testing and commissioning activities.
- Worked with industrial communication protocols including Modbus TCP and Ethernet-based communication.
- Supported automation system configuration and troubleshooting across PLC, HMI and SCADA environments.

TECHNICAL SKILLS

PLC Platforms

Siemens, Allen-Bradley, Schneider, ABB, Mitsubishi, Delta, Omron

SCADA / HMI

Wonderware InTouch, Citect SCADA, Vijeo Designer, Delta DOPSoft

Communication

Modbus TCP/IP, Modbus RTU, Ethernet/IP, ASCII, RS-232, DF1, Modbus Serial

Automation

Ladder Logic, PLC Programming, HMI Configuration, VFD Parameter Configuration, Troubleshooting, Testing & Commissioning

Programming

Python, C, Ladder Logic, Linux Basics

Networking

TCP/IP, IP Addressing, Ethernet Switches, Basic Network Troubleshooting

Tools

Arduino IDE, Arduino IoT Cloud, Microsoft Office

PROJECTS

Metal Sorting Machine

Allen-Bradley MicroLogix 1400 PLC

- Designed and implemented an automated metal sorting system using an Allen-Bradley MicroLogix 1400 PLC.
- Developed ladder logic for automated material detection and sorting operations.
- Configured serial communication between PLC and external devices using RS-232 with DF1 protocol.
- Integrated capacitive and proximity sensors to detect and differentiate metallic and non-metallic items.

Automatic Mains Failure (AMF)

Allen-Bradley MicroLogix 1400 PLC

- Designed and developed a miniature Automatic Mains Failure system using an Allen-Bradley MicroLogix 1400 PLC.
- Implemented relay logic and 24V DC switching for automated mains/generator control.
- Used a bus coupler for load sharing and integrated control components for reliable switching.
- Interfaced the system with Wonderware InTouch SCADA for real-time monitoring and data logging.

Batching Process Automation

Siemens S7-1200 PLC / TIA Portal / WinCC Unified

- Developed an automated batching process using a Siemens S7-1200 DC/DC/DC PLC and ladder logic in TIA Portal.
- Implemented weight-based batching of city water, QR and KM products into the main mixing tank according to predefined proportions.
- Programmed a 3-minute agitator operation after achieving the required batch proportion before transferring material to filling lines.
- Designed WinCC Unified HMI for real-time monitoring, process visualization, alarm indication and data logging.

Controller-to-Controller Communication

Schneider M340 PAC / Modicon M221

- Configured communication between Schneider PAC M340 and Modicon M221 PLC using READ_VAR and WRITE_VAR function blocks.
- Configured PAC M340 as the Master and Modicon M221 as the Slave controller.
- Established Master-Slave communication over Modbus TCP/IP for controller-to-controller data exchange.

Smart Fuel System Using IoT

IoT / QR Payment

- Developed an IoT-based smart fuel system enabling real-time communication between users and petrol pump operations.
- Implemented a QR-based payment interface to improve transaction transparency and reduce customer waiting time.
- Focused on improving system security, user convenience and interaction between the user and fuel dispensing system.

Image Denoising Using Machine Learning

Python / TensorFlow / OpenCV / PyTorch

- Developed machine learning-based image denoising solutions for Gaussian, Salt & Pepper, Speckle and Uniform noise.
- Built an end-to-end image denoising web application using Django and OpenCV, enabling noisy image upload and side-by-side restoration results.
- Designed a CNN-based noise classification pipeline to classify five noise types: Gaussian, Salt & Pepper, Poisson, Speckle and Uniform.
- Implemented a Patch-Based U-Net architecture using TensorFlow/Keras for high-resolution image restoration while retaining spatial features and edges.
- Integrated automated quality metrics including PSNR, SSIM, FSIM and NPS for quantitative evaluation, achieving up to 16.61 dB PSNR improvement.

Electronic Medicine Dispenser

ESP32 / C/C++ / Arduino IoT Cloud

- Designed and developed an automated medicine dispenser using ESP32 for scheduled dispensing of solid pills and liquid dosages.
- Integrated Arduino IoT Cloud and Wi-Fi for remote medication monitoring, user alerts and real-time telemetry.
- Implemented compliance tracking using IR sensors and piezo buzzers to verify intake and trigger missed-dose alerts.
- Interfaced servo motors, water pump motor and flow-rate sensor using C/C++ in Arduino IDE.

CERTIFICATIONS

- **Professional Diploma in Industrial Automation (PDIA)** – IPCS Global — PLC Programming, SCADA, HMI, VFD, DCS
- **Computer Architecture** – NPTEL, IIT Madras
- **Embedded Systems in EV** – Techmaghi
- **CSRBOX Placement Assistant Program** – IBM SkillsBuild

LEADERSHIP & VOLUNTEERING

- **Placement Coordinator**, ECE Department, Government Engineering College, Palakkad
- **Department Program Coordinator**, ECE Department, Government Engineering College, Palakkad